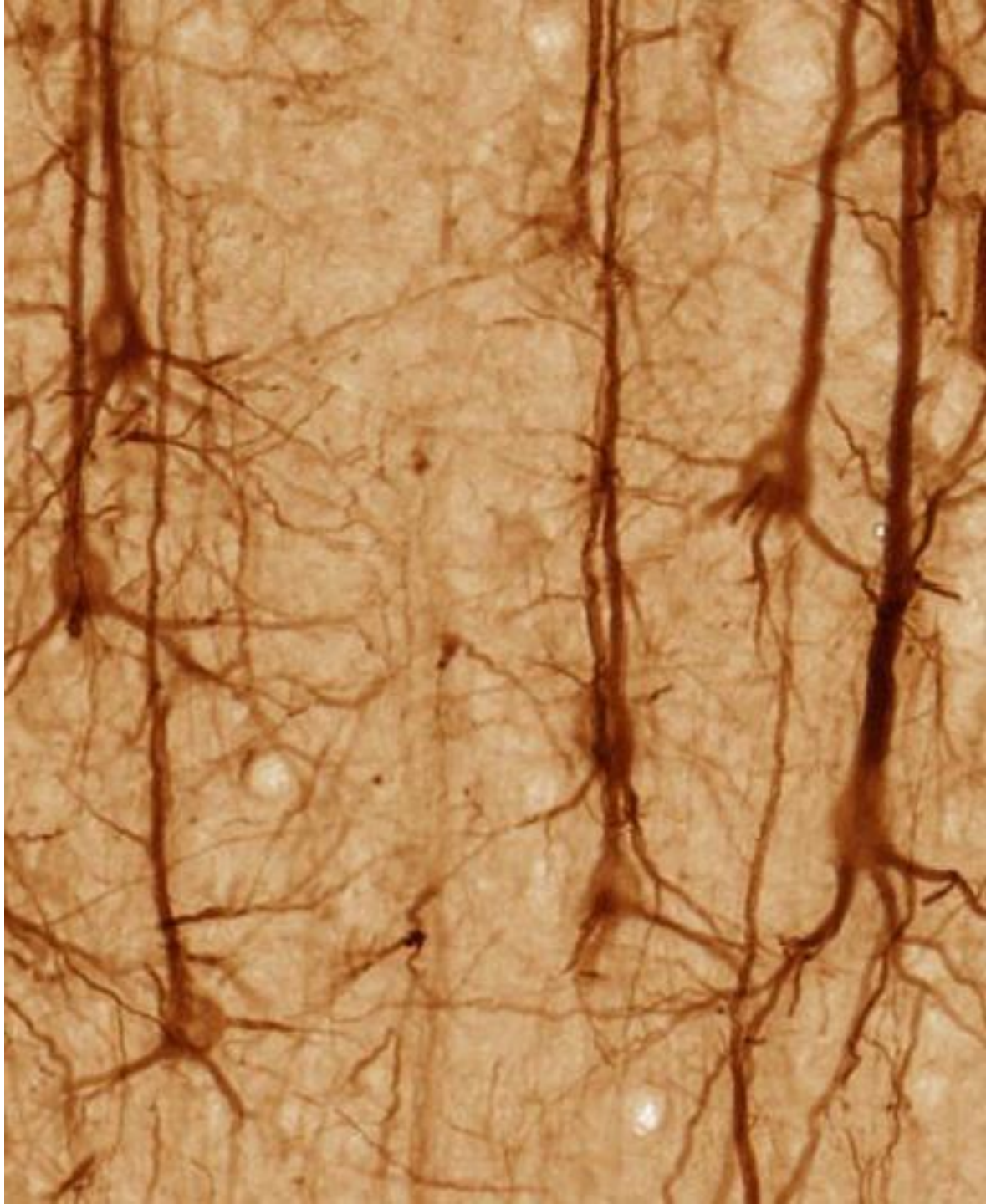
DP Biology – C2.2

Neural signalling

- How do neurons produce electrical signals and pass them along?
- How do neurons affect other cells?

Statements C2.2.1–7, SL and HL. Textbook pp. 467–478.
Image: stained neurons, light micrograph. UC Regents Davis campus, CC BY 3.0.

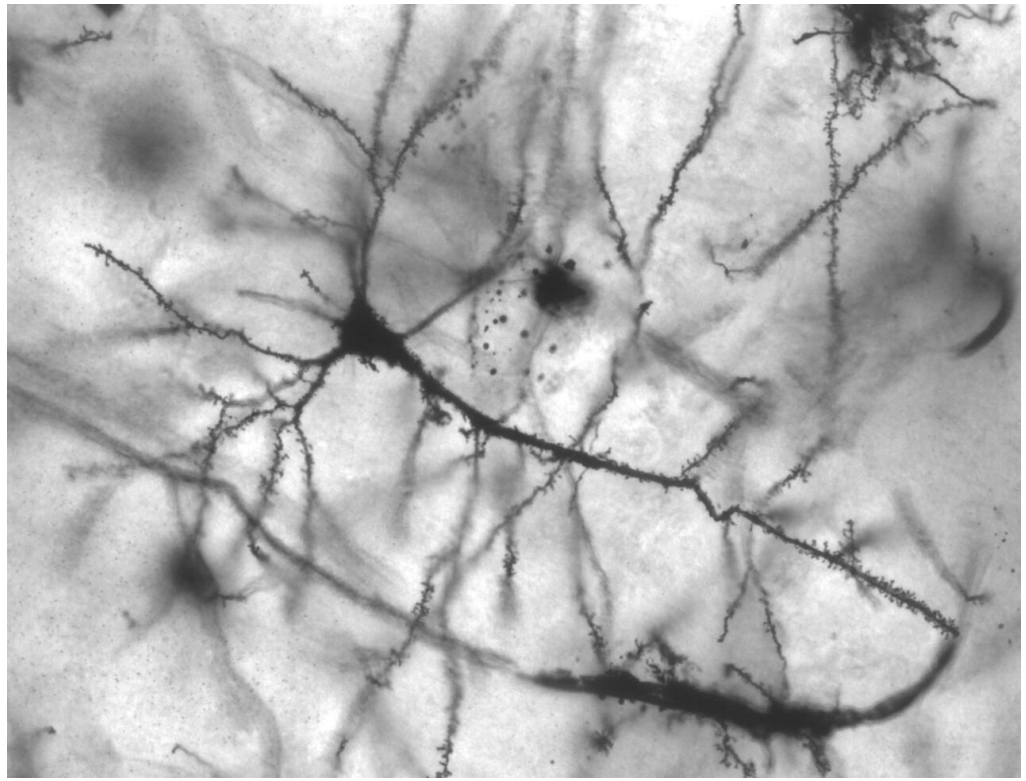


Objectives for each class

Class	Statements	By the end of the class, you can…
1	C2.2.1–2	Describe the parts of a neuron, and explain how pumping and leakage set up the resting potential
2	C2.2.3–4	Describe an action potential, and explain the effects of axon diameter and myelin on speed
3	C2.2.5–7	Explain how a signal crosses a synapse, using acetylcholine as the example

Neurons carry electrical signals

C2.2.1



A neuron from the brain, stained. MethoxyRoxy, CC BY-SA 2.5.

The **central nervous system** (CNS) is the brain and spinal cord. **Nerves**, bundles of nerve fibres, link it to every part of the body.

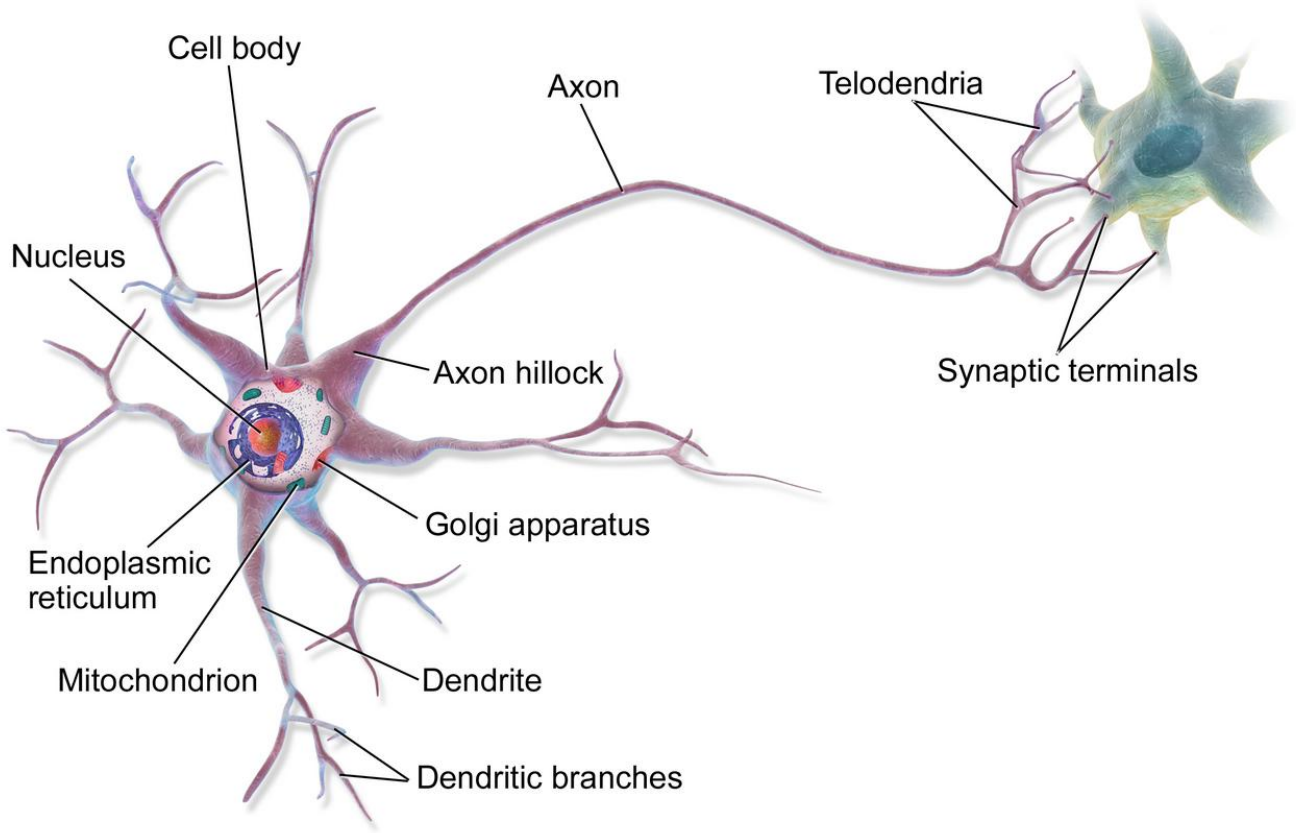
Neurons are the cells that carry **electrical impulses**: fast signals sent to particular places.

Link

Hormones also coordinate the body, but they travel in the blood and act more slowly (C2.1, C3.1).

The structure of a neuron

C2.2.1



A neuron and its parts. BruceBlaus, CC BY 3.0.

Cell body — nucleus and most of the cytoplasm

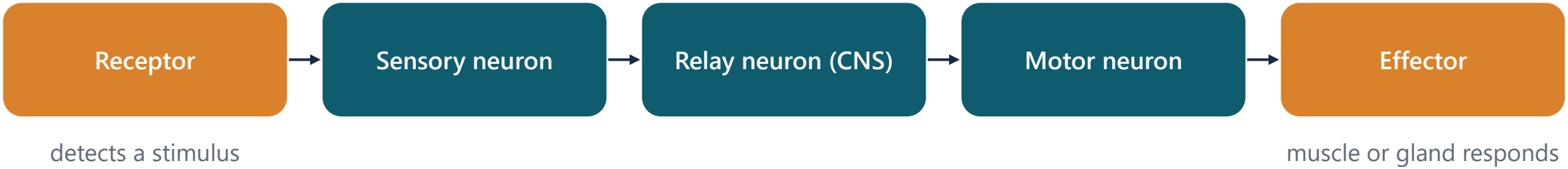
Dendrites — many short fibres: bring impulses in

Axon — one long fibre: carries impulses away

direction of the impulse: dendrites → cell body → axon

Neurons in a pathway

C2.2.1



Sensory neurons carry impulses to the CNS; **relay** neurons link neurons inside it; **motor** neurons carry impulses to effectors.

Common mistake

Receptors detect; effectors respond. Do not mix them up.

A resting neuron is polarized

C2.2.2



A charge difference across a membrane is a **membrane potential**. A membrane with one side positive and the other negative is **polarized**.

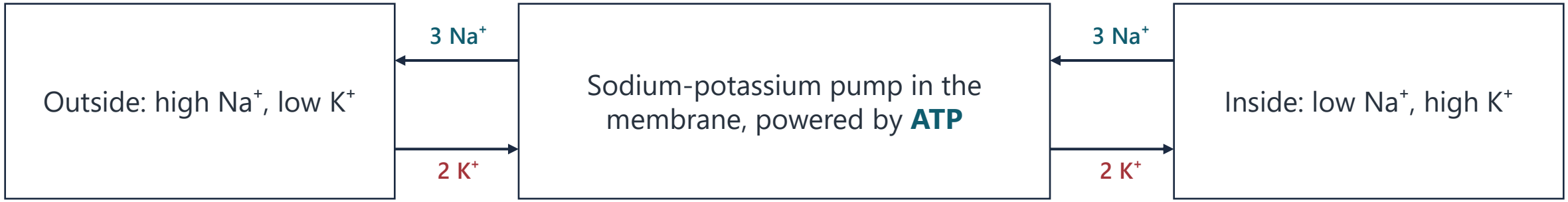
In a resting neuron the inside is about **−70 mV** compared with the outside: the **resting potential**.

Key idea

A resting neuron keeps using ATP to hold this difference, ready for the next impulse.

The sodium-potassium pump

C2.2.2



Active transport: 3 Na^+ out and 2 K^+ in for each ATP, against their gradients. This builds **two concentration gradients** and moves more positive charge out than in.

Key idea

ATP comes from aerobic respiration, so nerves need a steady supply of oxygen and glucose even at rest.

Why the resting potential is negative

C2.2.2

Cause	Effect
Sodium-potassium pump	More positive ions moved out than in
Leakage	K^+ leaks out faster than Na^+ leaks in: the membrane is more permeable to K^+
Proteins	Large negatively charged proteins stay inside

Exam note

Be ready to give the energy source (ATP), the meaning of polarized, and why the value is negative. Remember about −70 mV, with its sign.

Practice: ions across the membrane

C2.2.2

Typical ion concentrations in a mammalian neuron, in mmol dm^{-3}.

Ion	Inside	Outside
Na^+	15	145
K^+	140	5

1. How many times higher is Na^+ outside than inside?
2. If channels opened, which way would each ion diffuse?
3. Explain why the pump needs ATP.
4. Cyanide stops aerobic respiration. Predict what happens to the resting potential.

Check-in: class 1

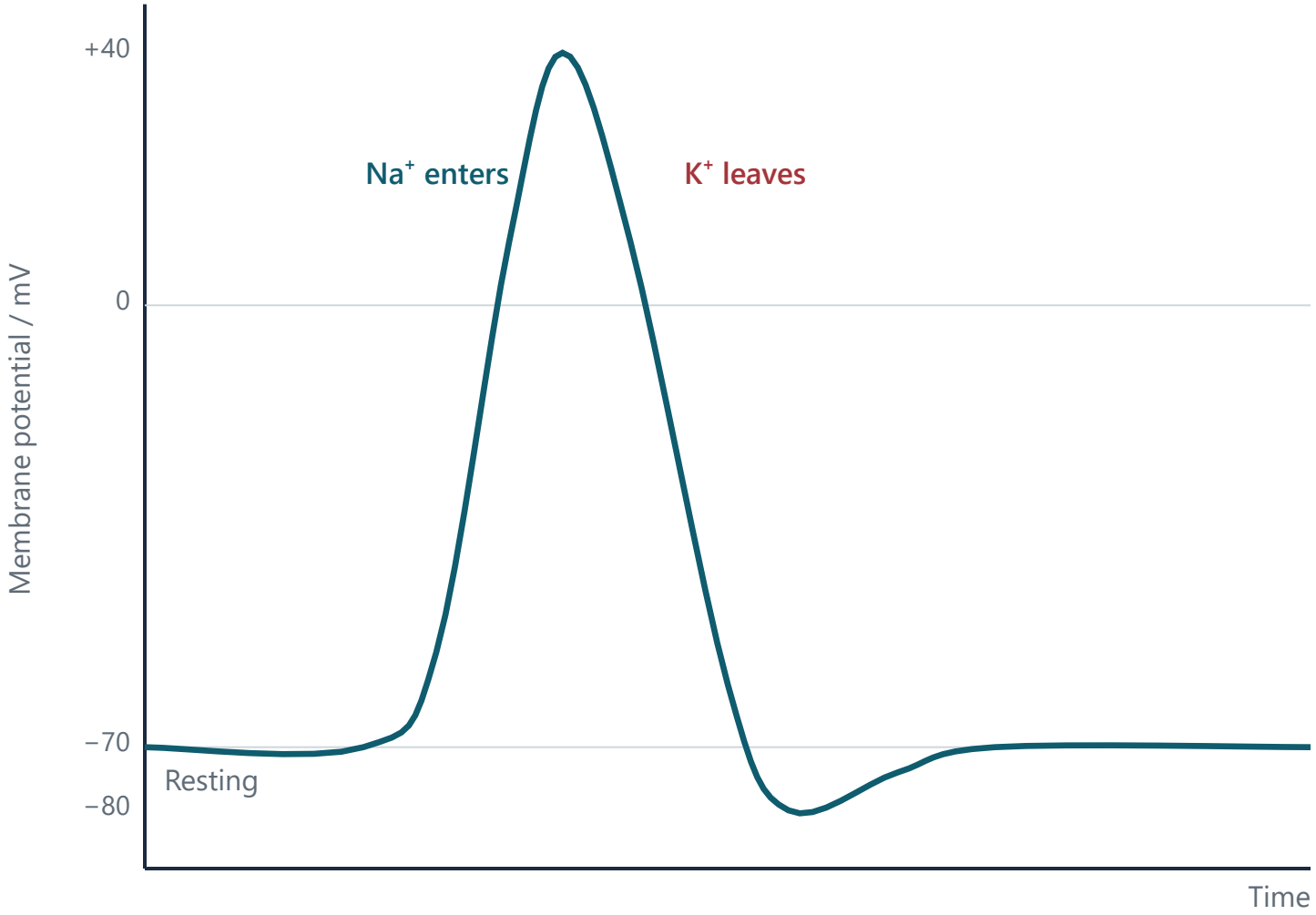
C2.2.1–2

1. Distinguish between dendrites and an axon.
2. State the roles of sensory, relay and motor neurons.
3. Define resting potential and give its typical value.
4. Describe how the sodium-potassium pump works.
5. Give three reasons why the inside of a resting axon is negative.

How to use this

Answer without your notes, then check against slides 3–9.

An action potential



Schematic, not to scale. One point on the axon over a few milliseconds.

A stimulus lets **Na^+ flood in**: the potential reverses from −70 to about **+40 mV**. This is **depolarization**.

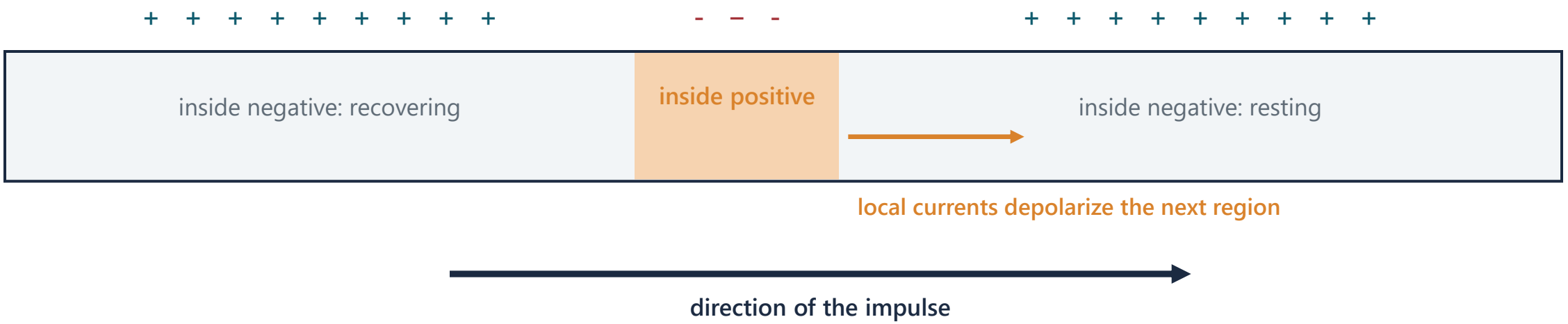
Then **K^+ flows out** and the inside becomes negative again: **repolarization**.

HL

Channels and thresholds are in C2.2.8–10.

The impulse travels along the axon

C2.2.3



A **nerve impulse** is an action potential that moves along the fibre. Each point is depolarized for about 2 ms, which triggers the next point, and so on.

It is **electrical** because charged ions move across the membrane, changing the voltage.

Think

Why is an impulse not like a current flowing along a copper wire?

Wider fibres conduct faster

C2.2.4



Squid (Loligo). Hans Hillewaert, CC BY-SA 4.0.

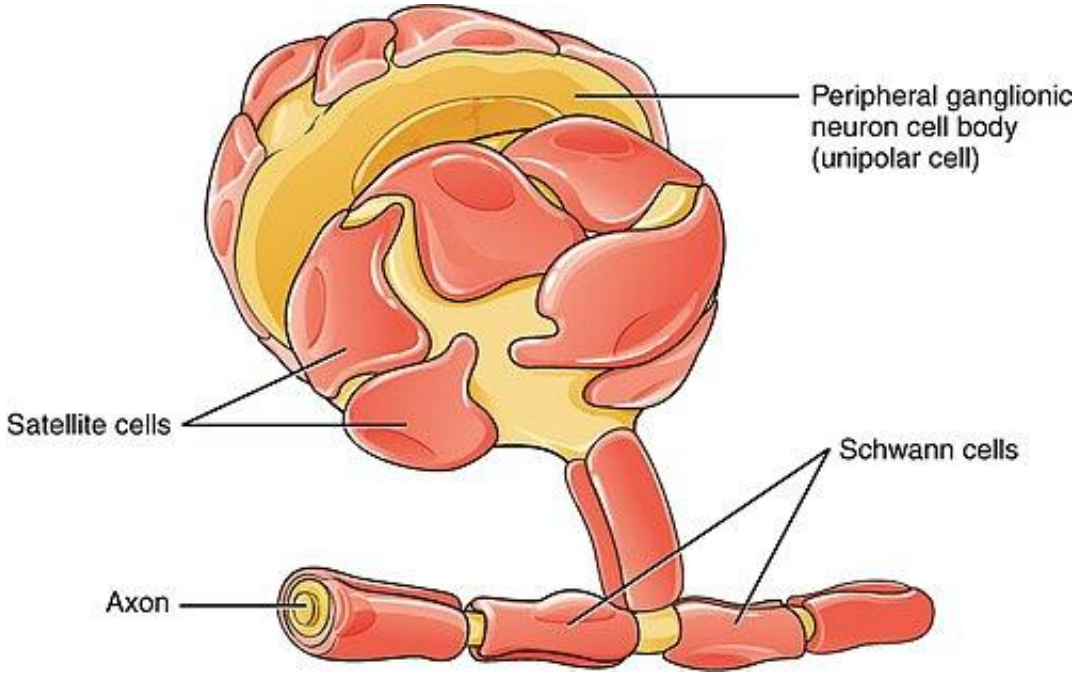
A narrow fibre has more **resistance** to the flow of ions, so impulses are slower.

Small fibres, about 1 µm wide: about 1 m s^{-1}. **Squid giant axons**, up to 500 µm: about 25 m s^{-1}.

Think

Suggest how a giant axon helps a squid survive, and why humans cannot have giant axons in all their nerves.

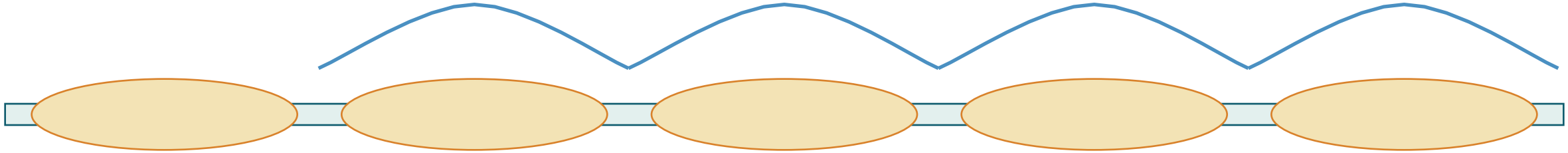
Myelin greatly increases speed



Schwann cells wrap around axons. OpenStax, CC BY 4.0.

Schwann cells wrap the axon in a fatty **myelin sheath**, which insulates it.

The membrane is exposed only at gaps, the **nodes of Ranvier**. The impulse jumps from node to node: up to about **100 m s^{-1}**.



myelin sheath (yellow) with nodes between; blue: the impulse jumps from node to node

Comparing conduction speeds

C2.2.4

Fibre	Diameter	Myelinated?	Typical speed
Small fibre in a mammal	1 μm	No	about 1 m s^{-1}
Squid giant axon	500 μm	No	about 25 m s^{-1}
Mammal motor fibre	20 μm	Yes	about 100 m s^{-1}

Activity

Monash University simulation: ilearn.med.monash.edu.au/physiology/action-potentials/axon-diameter
Run it for non-myelinated, then myelinated, fibres. Plot speed against diameter for both and describe the difference.

Working with data: diameter, myelin and speed

C2.2.4

Conduction speeds in different nerve fibres (approximate published values).

Fibre	Diameter / μm	Myelin?	Speed / m s^{-1}
Squid giant axon	500	No	25
Human pain fibre	1	No	1
Human touch fibre	10	Yes	50
Human motor fibre	15	Yes	100

1. Compare the squid giant axon with the human motor fibre.
2. Explain how a narrow fibre can be faster than the squid axon.
3. Calculate how long a motor impulse takes to travel 1 m.
4. Suggest why slow pain fibres give a dull, lasting pain.

Check-in: class 2

C2.2.3–4

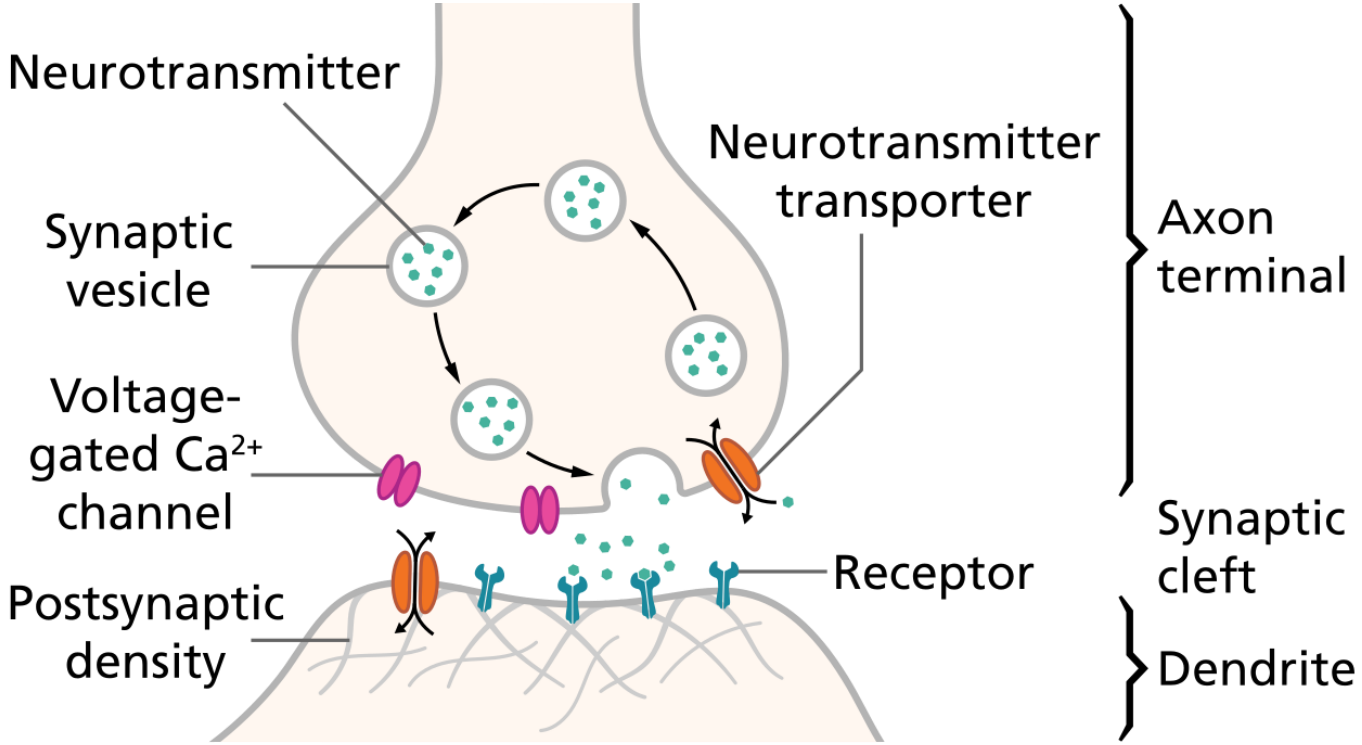
1. Describe what happens in depolarization and repolarization.
2. Explain how an action potential moves along an axon.
3. Explain why a nerve impulse is described as electrical.
4. Explain the effect of axon diameter on speed.
5. Explain how myelin increases the speed of impulses.

How to use this

Answer without your notes, then check against slides 11–16.

Synapses are junctions

C2.2.5



A chemical synapse. Thomas Splettstoesser, CC BY-SA 4.0.

The **presynaptic** neuron ends in a swollen knob. The **postsynaptic** cell receives the signal.

A **synaptic cleft** about 20 nm wide separates them. An action potential cannot cross it.

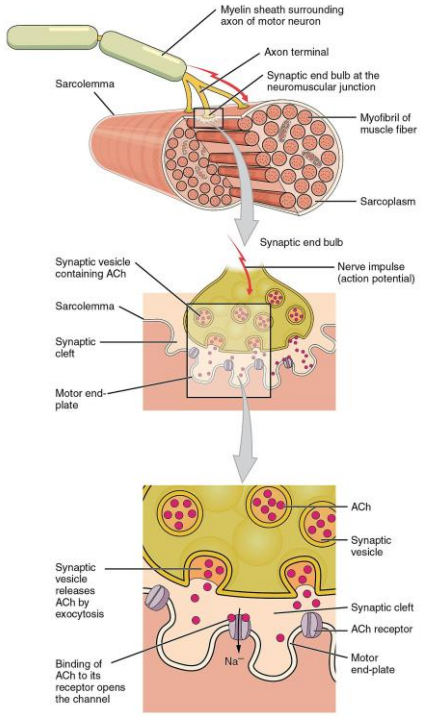
Key idea

Chemicals called **neurotransmitters**, stored in vesicles, carry the signal across.

Where synapses are found

C2.2.5

Junction	Example
Neuron to neuron	Brain and spinal cord
Neuron to muscle	Neuromuscular junction
Neuron to gland	Salivary gland



Neuromuscular junction. OpenStax, CC BY 4.0.

One direction only

Vesicles of neurotransmitter are only in the presynaptic knob; receptors are only on the postsynaptic membrane.

Releasing the neurotransmitter

C2.2.6

Action potential reaches the knob

↓

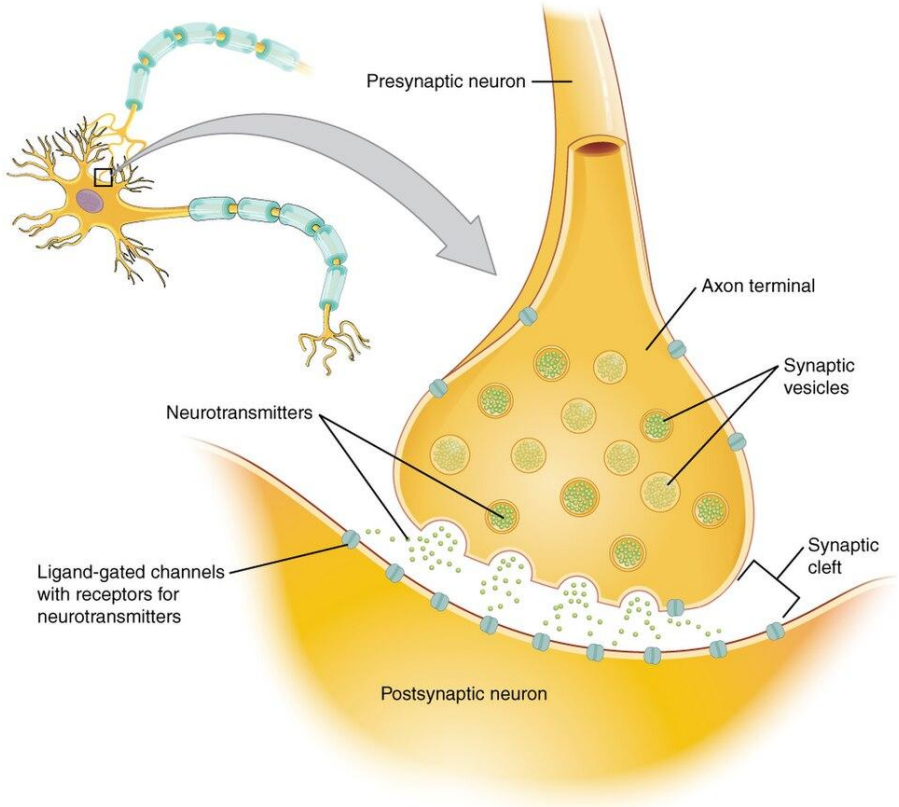
Ca^{2+} channels open; Ca^{2+} diffuses in

↓

Vesicles move to the membrane and fuse

↓

Neurotransmitter released into the cleft



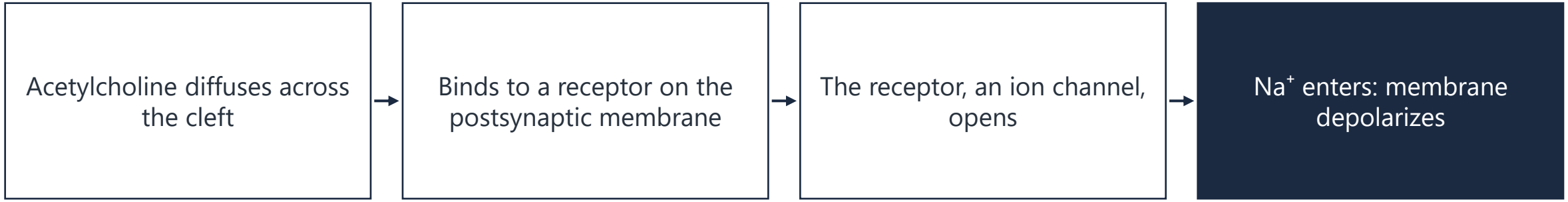
Events at a chemical synapse. Young et al., OpenStax, CC BY 4.0.

Key idea

Calcium acts as a signal inside the knob. Electrical in, chemical across, electrical again.

An excitatory postsynaptic potential

C2.2.7



This depolarization is an **excitatory postsynaptic potential**. If it is large enough, an action potential starts in the next cell.

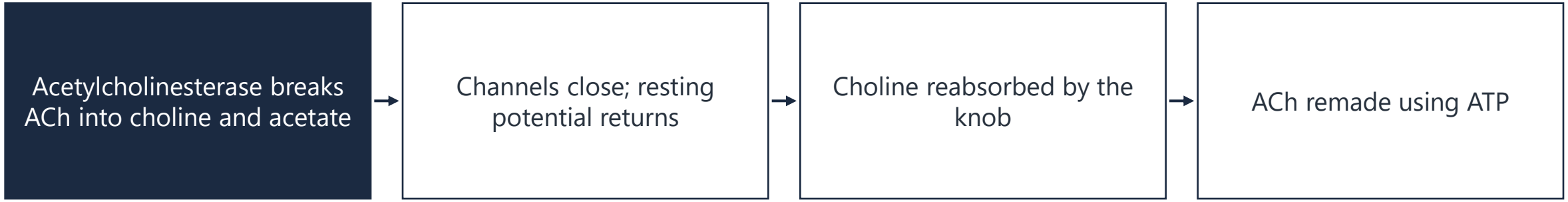
Use **acetylcholine (ACh)** as your example: it acts at many synapses, including the neuromuscular junction.

Common mistake

Acetylcholine does not "move" or get pumped across the cleft. It diffuses, from high to low concentration.

Clearing the cleft

C2.2.7



If ACh stayed in the cleft, the postsynaptic cell would keep being stimulated.

Activity

Draw the whole synapse sequence as one numbered flow chart, from the action potential arriving to ACh being remade. Cover it and redraw it from memory. Then explain why the knob has many mitochondria.

Practice: synapse problems

C2.2.6–7

Put in order

(A) Na^+ enters the next cell. (B) Ca^{2+} enters the knob. (C) ACh diffuses across. (D) Vesicles fuse. (E) ACh binds receptors.

Predict

An insecticide blocks acetylcholinesterase. What happens at the insect's muscles?

Explain

A drug blocks ACh receptors. Why does the muscle stop responding?

How to use this

Answer each in two or three sentences, using the key words from slides 18–22.

Check-in: class 3

C2.2.5–7

1. Define a synapse.
2. Explain why a synapse works in one direction only.
3. Outline the role of calcium ions at the presynaptic membrane.
4. Describe how ACh produces an excitatory postsynaptic potential.
5. Explain why ACh must be removed from the cleft.

How to use this

Answer without your notes, then check against slides 18–23.

Common mistakes to avoid

Not this	But this
An impulse is a current flowing along the axon.	It is a wave of depolarization: ions cross the membrane at each point.
Na^+ and K^+ pass through the pump during an action potential.	They diffuse through channels; the pump restores the gradients.
Myelin conducts the impulse.	Myelin insulates; the impulse jumps between nodes.
Neurotransmitter is pumped across the cleft.	It diffuses across the cleft.
The resting potential is 0 mV.	It is about −70 mV: the inside is negative.

Summary: neural signalling

C2.2.1–7

	Key idea
C2.2.1	Neurons: cell body, dendrites in, axon out
C2.2.2	Pump (3 Na^+ out, 2 K^+ in) and leakage give −70 mV
C2.2.3	Action potential: Na^+ in, then K^+ out; it moves along the fibre
C2.2.4	Wider fibres and myelin give faster impulses
C2.2.5	Synapses link neurons to neurons, muscles and glands
C2.2.6–7	Ca^{2+} in, vesicles release ACh, Na^+ enters the next cell

Check that you can...

1. **Describe** the structure of a neuron
2. **Explain** how the resting potential is set up and maintained
3. **Describe** an action potential as depolarization and repolarization
4. **Explain** how diameter and myelin affect speed
5. **Outline** where synapses are found
6. **Explain** how ACh is released and produces an excitatory postsynaptic potential

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Give two reasons why the inside of a resting axon is negative.
2. Explain how myelin increases the speed of a nerve impulse.
3. Explain why a signal can cross a typical synapse in only one direction.

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